



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Dept. of Computer
Science Faculty of
Science and Technology**

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Fall 2024-2025

Section: [k] Group No: 02

Project Report On

PROJECT NAME : Day-Care Management System

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2. MD. MOZAHIDUL ISLAM	23-51293-1	Child Update all, Staff Update Control, Shop management with 3 tab , Payments , Report
3.ESHFAQ JAHAN CHHOYA	23-50659-1	Packages Function , Payments , Parents
4.DIPU ROY	23-50420-1	Normalization , Front Form design, Sql

INTRODUCTION :

The Day Care Management System, developed in C# with the .NET Framework and powered by MS SQL Server, is a modern solution tailored to improve the efficiency of childcare operations. This system stitches together key stakeholders such as administrators, parents, staff, and children for easy management of essential tasks: child profile updates, recruitment of staff, processing of payments, activity tracking, and many more.

It has been designed in such a way that it is scalable and organized, hence implementing database normalization to ensure no redundancy or loss of integrity of the data. The platform simplifies operations in managing daily activities down to updating staff and child records, hence fostering transparency and communication among all stakeholders involved. This system ensures a well-organized, secure, and caring environment for children, parents, and staff with a centralized, user-friendly interface.

PROJECT DESCRIPTION :

The Day Care Management System is a robust database-driven application developed to manage child care operations in an organized and systematic manner. It is designed with C# using the .NET Framework and powered by MS SQL Server, including admins, parents, staff, children, and financial operations. The project deals basically with the issues of unstructured data, applying rigor for normalization for clarity, no redundancy, and integrity.

System Features :

- ❖ Admins to manage overall functionality of the system, adding recruits, child profiling, and other processes with full security and update options.
- ❖ The parents shall register themselves, amend the child information, track the activities, and pay through this system.
- ❖ Creating staff profiles and updating upon hiring, updating credentials, and keeping current with their records of responsibilities.
- ❖ Religious tracking of the activities and updates of the attendance, health records, and schedules of children and associating those to the parent's and staff profile.

It also allows for efficient payment processing, subscriptions, and fees, which can be in line with the needs of the parents: monthly, yearly, or even custom plans. It has a kids' activity management module

that helps in managing scheduling and updates, hence making operations smooth.

The database design is based on normalization, moving from Un-Normalized Form (UNF) to First, Second, and Third Normal Forms (1NF, 2NF, 3NF). This ensures that each piece of data logically relates to its primary key, eliminates transitive dependencies, and normalizes data into atomic and non-redundant components. "Oversees," "Updates," "Manages," "Processes," and "Tracks" are some of the relationships that are well captured in the database to ensure smooth interaction among all entities.

The final system, as envisioned in its ER diagram, is scalable and well-structured in database architecture, which can respond to the complex needs of a childcare management facility. This project really shows the power of normalization of databases in changing fragmented, unorganized data into an efficient and user-friendly platform that enables smooth operations and ensures the best care for children.

ER DIAGRAM :

A daycare management system is designed to computerize the administrative and managerial operations of a daycare center. It helps in handling crucial operations like maintaining children's records, staff assignments, parents' details, and administrative concerns.

This ER diagram graphically represents the structure of the database for a Daycare Management System, showing the relationships among key entities. The diagram includes:

Parents - The table stores details such as ID, name, contact information, address, and profession.

Children: This would involve a child's ID, name, date of birth, guardian's details, medical records, and so on.

Staff: This involves information about staff members, including job titles, salaries, and contact addresses.

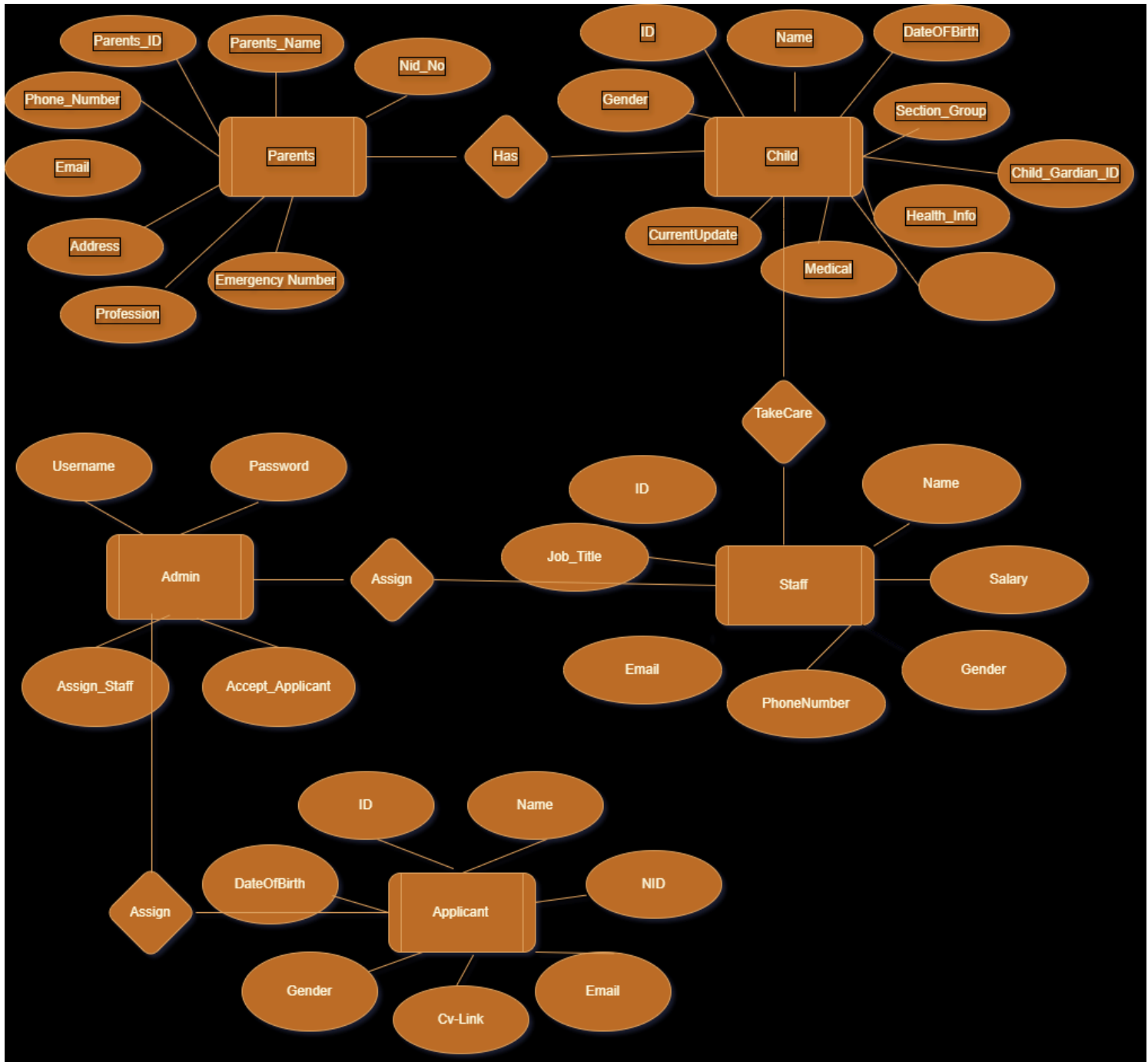
Admin: This controls the operation of daycare, staff assignment to accept applicants.

Applicants: This stands for the person applying to work in your daycare or prospective staff; it shows links to their CVs, displaying personal information.

Each entity is related to others through "Has," "TakeCare," and "Assign" relationships for smooth interaction between different components of the system.

This ER diagram gives a clear structure for developing a database that efficiently supports daycare management, ensuring better organization, security, and accessibility of data.

Daycare Management System



Admins Managing Staff :

1NF -

AdminID, Password, StaffID, StaffName, StaffRole.

2NF -

Admin Table: AdminID (PK), Password

Staff Table: StaffID (PK), StaffName, StaffRole

Admin-Staff Relationship: AdminID (FK), StaffID (FK)

Staff Handling Children :

1NF -

StaffID, StaffName, ChildID, ChildName, DateOfBirth

2NF -

Staff Table: StaffID (PK), StaffName

Child Table: ChildID (PK), ChildName, DateOfBirth

Staff-Child Relationship: StaffID (FK), ChildID (FK)

Parents Registering Children :

1NF -

ParentID, ParentName, Email, ChildID, ChildName, RegistrationDate

2NF –

Parent Table: ParentID (PK), ParentName, Email

Child Table: ChildID (PK), ChildName

Parent-Child Relationship: ParentID (FK), ChildID (FK), RegistrationDate

Updating Child Details :

1NF-

ChildID, ChildName, DateOfBirth, UpdatedBy (StaffID), UpdateDate

2NF –

Child Table: ChildID (PK), ChildName, DateOfBirth

Staff Table: StaffID (PK)

Update Table: ChildID (FK), StaffID (FK), UpdateDate

Final Normalized Tables (3NF) :

Admin Table: AdminID (PK), Password

Staff Table: StaffID (PK), StaffName, StaffRole

Admin-Staff Relationship: AdminID (FK), StaffID (FK)

Child Table: ChildID (PK), ChildName, DateOfBirth

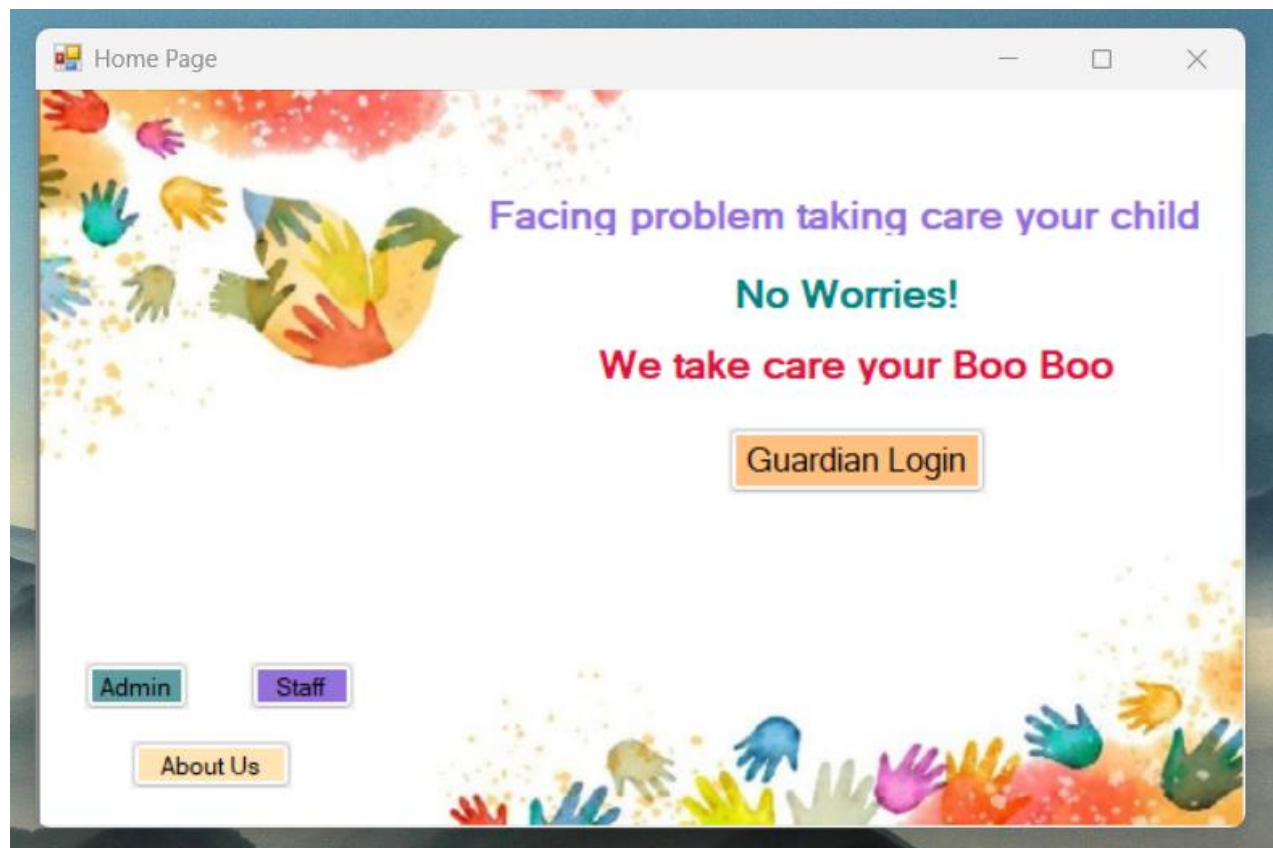
Staff-Child Relationship: StaffID (FK), ChildID (FK)

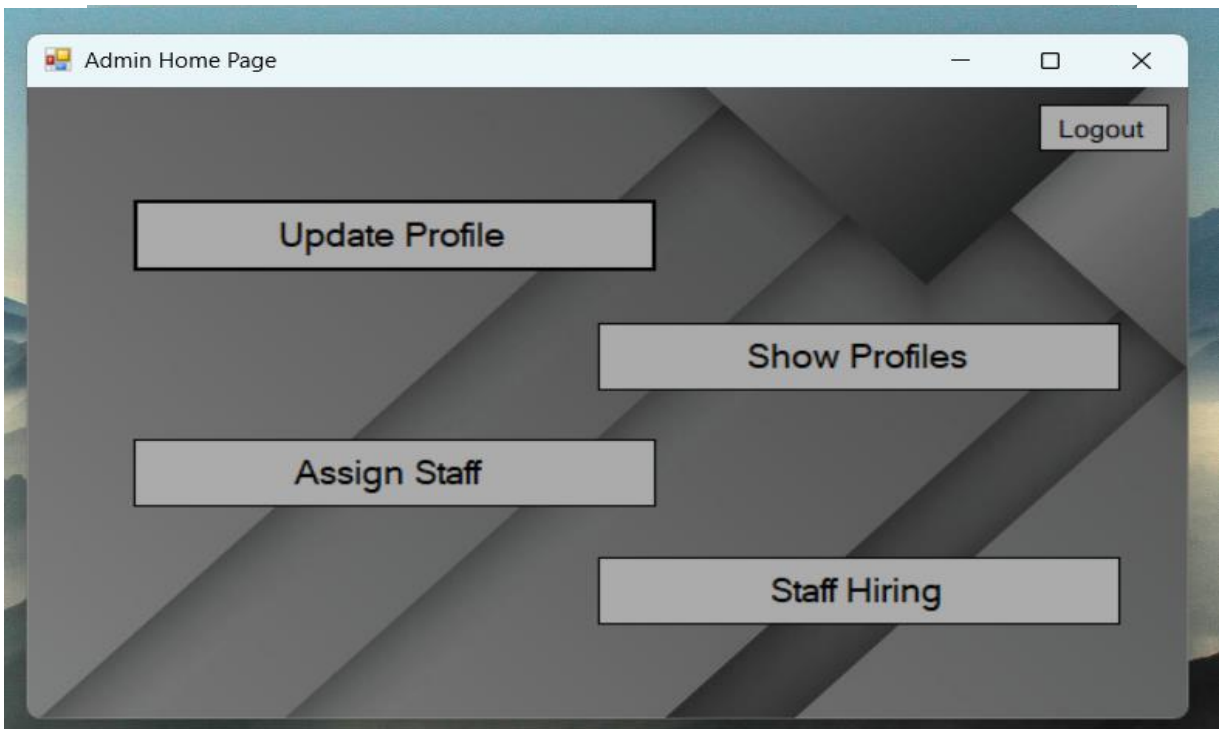
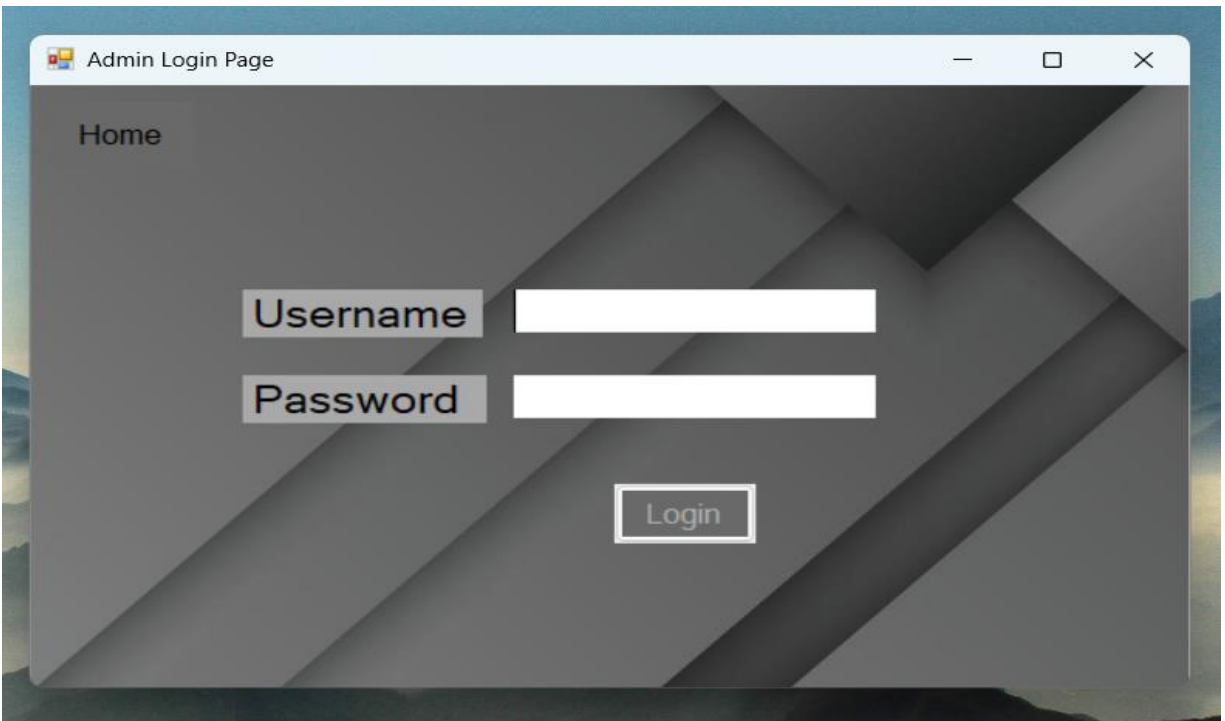
Parent Table: ParentID (PK), ParentName, Email

Parent-Child Relationship: ParentID (FK), ChildID (FK), RegistrationDate

Update Table: ChildID (FK), StaffID (FK), UpdateDate

USER INTERFACE :





All Profiles

Guardian

Child

Staff

Guardian Data

Refresh

Back

Search

guardian_id

guardian_l_id

guardian_l_name

guardian_l_na

guardian_l

1002

1234

Yasser

3456456

1005

234

Begum

112323

1006

112

Shan

227712

1007

dh843cb48

Shahman

3683345

1010

gjei496hg90

Kamal

Hasan

3580643

1013

1111

Shamin

Yasser

948356

Assign Staff

Back

staff_id

staff_password

staff_f_name

staff_l

3001

123

Salma

Begum

3002

453

Nita

Ahmed

3003

229

Kamal

Mia

3004

107

Dipa

Saha

3005

123456789

Akkas

Rahma

*

child_id

child_f_name

child_l_name

child_l

2004

Alia

Bhatun

20

2005

zzz

zzza

aa

2006

tf

saf

ds

*

Assign

Staff Hire

ID

☐ Nursery ☐ Toddler ☐ PreSchool

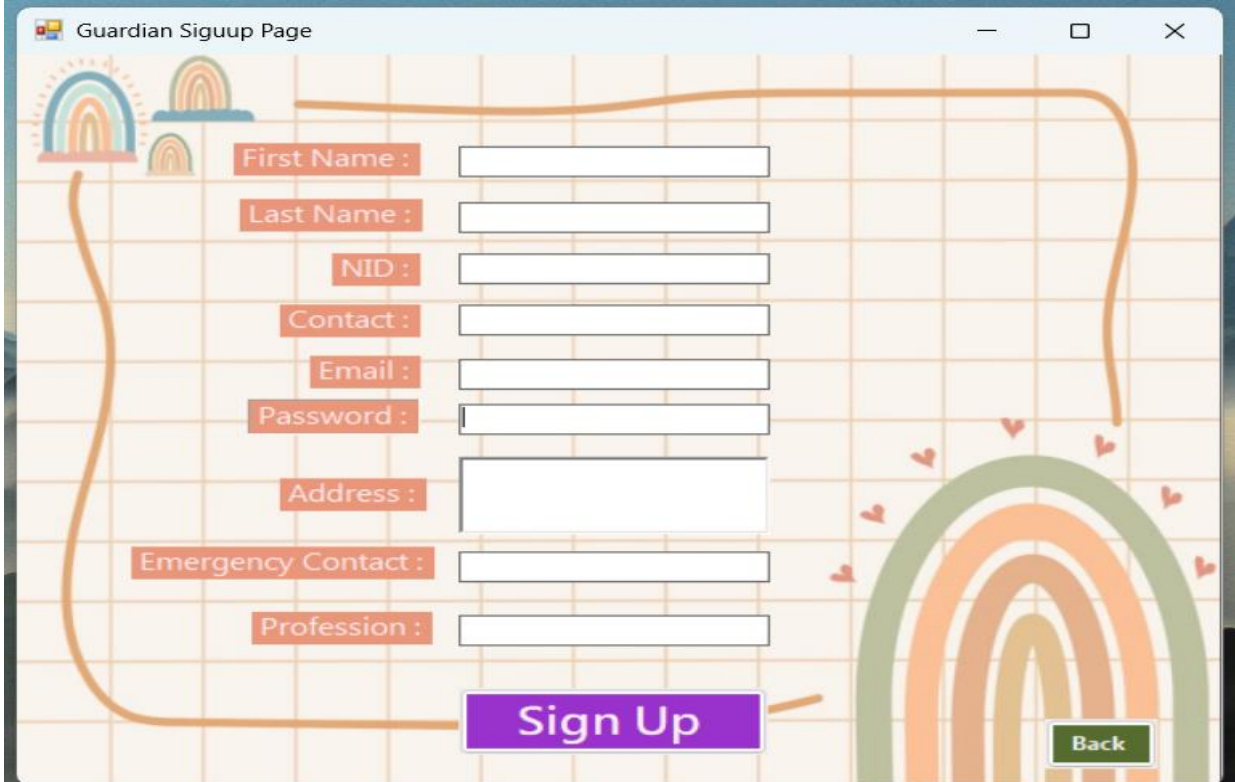
	applicant_id	applicant_f_name	applicant_l_name	applicant_nid	applicant_dob	appli
▶	4013	aaa	aaa	aa	Thursday, Januar...	Fema

Guardian Login Page

ID :

Password :

Guardian Signup Page



The page features a light orange grid background. On the left, there are three small rainbow icons. On the right, there is a large, multi-colored rainbow with small red hearts floating around it. A thick orange line starts from the top left, loops around the form fields, and ends at the 'Sign Up' button. The form fields are arranged vertically and each has a label in a red box to its left.

First Name :

Last Name :

NID :

Contact :

Email :

Password :

Address :

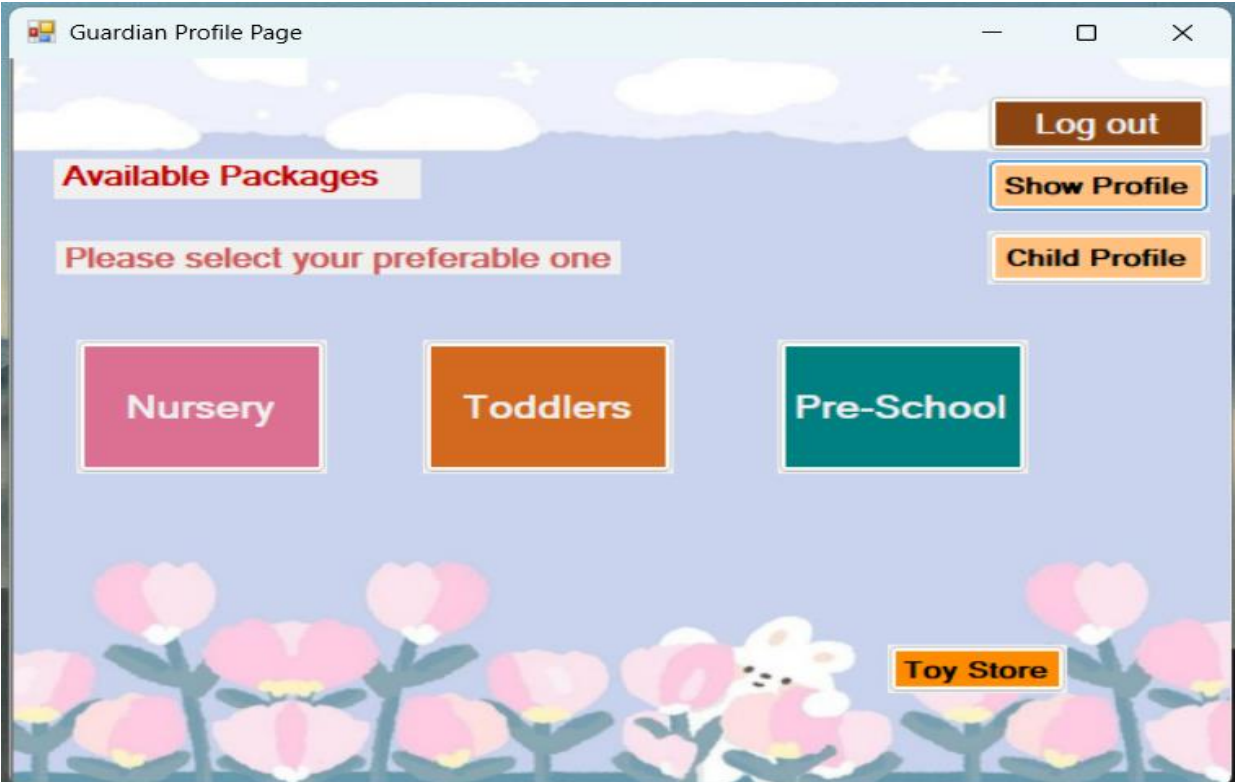
Emergency Contact :

Profession :

[Sign Up](#)

[Back](#)

Guardian Profile Page



The page has a light blue background with a sky and clouds illustration at the top and a row of pink flowers at the bottom. On the right side, there are three buttons: 'Log out' (brown), 'Show Profile' (orange), and 'Child Profile' (orange). On the left, there is a section titled 'Available Packages' in red, followed by the instruction 'Please select your preferable one' in red. Below this, there are three colored boxes: 'Nursery' (pink), 'Toddlers' (orange), and 'Pre-School' (teal). At the bottom right, there is a 'Toy Store' button (orange) next to a small white dog icon.

[Log out](#)

[Show Profile](#)

[Child Profile](#)

Available Packages

Please select your preferable one

[Nursery](#)

[Toddlers](#)

[Pre-School](#)

[Toy Store](#)

Guardian Profile Page

ID	1002
New First Name	Shamin
New Last Name	Yasser
New Nid	345645674321
New Contact	000000000
New Email	shaminyasser001@gmail.com
New Address	Bogra
Emergency Contact	01987654321
New Profession	Student
New Password	1234

Child Update Profile

First Name		Rahima
Last Name		Alam
DOB		2015
Blood Group		B+
Gender	☐ Male ☐ Female	Female
Health Issue		NULL

Nursery Details

—

□

×

Nursery

Safe and nurturing environment for infants.
Focus on sensory stimulation and early motor skill development.
Personalized care routines. Such as feeding, napping and so on.
Music and sound exploration.
Tummy time and physical development activities.
Gentle interactions and storytime.

☐ 1 Day - 700tk ☐ 7 Day - 4600tk ☐ 30 Day - 17500tk

Back

Payment

Toddlers Details

—

□

×

Toddlers

Safe and nurturing environment for infants.
Focus on sensory stimulation and early motor skill development.
Personalized care routines. Such as feeding, napping and so on.
Music and sound exploration.
Tummy time and physical development activities.
Gentle interactions and storytime.

☐ 1 Day - 750tk ☐ 7 Day - 5000tk ☐ 30 Day - 19000tk

Back

Payment

Pre-School Details

Pre-School

Safe and nurturing environment for infants.
Focus on sensory stimulation and early motor skill development.
Personalized care routines. Such as feeding, napping and so on.

Music and sound exploration.
Tummy time and physical development activities.
Gentle interactions and storytime.

1 Day - 800tk

7 Day - 5200tk

30 Day - 20000tk

Back

Payment

Staff_Input_Data

Go Back

Enter Child Data

Go To Home

Child ID		Incident	
Staff ID		Medical	
Solid		Need	
Diaper		Potty	
Sleep		Bottle	
Activity		Message	
Mood		Submit Info	Sign Out

Payments.cs [Design]Program.csSearchChild.cs [Design]Home_Page.cs [Design]

Back

Payment Details

Quantity

Payment Method

Card Details

Card Number

Expiry Date

CVV

CancelProceed to

Toy Store

GirlsBoysToys

Buy Your Kid's Product



Headband Tutu Skirt

Add Card 100



Short Sleeves

Add Card 100



Silver Flower Sleeves

Add Card 100



Embroidered Princess

Add Card 100



Grey and Silver overlay

Add Card 100



Silver Flower

Add Card 100



Artificial Bridesmaid

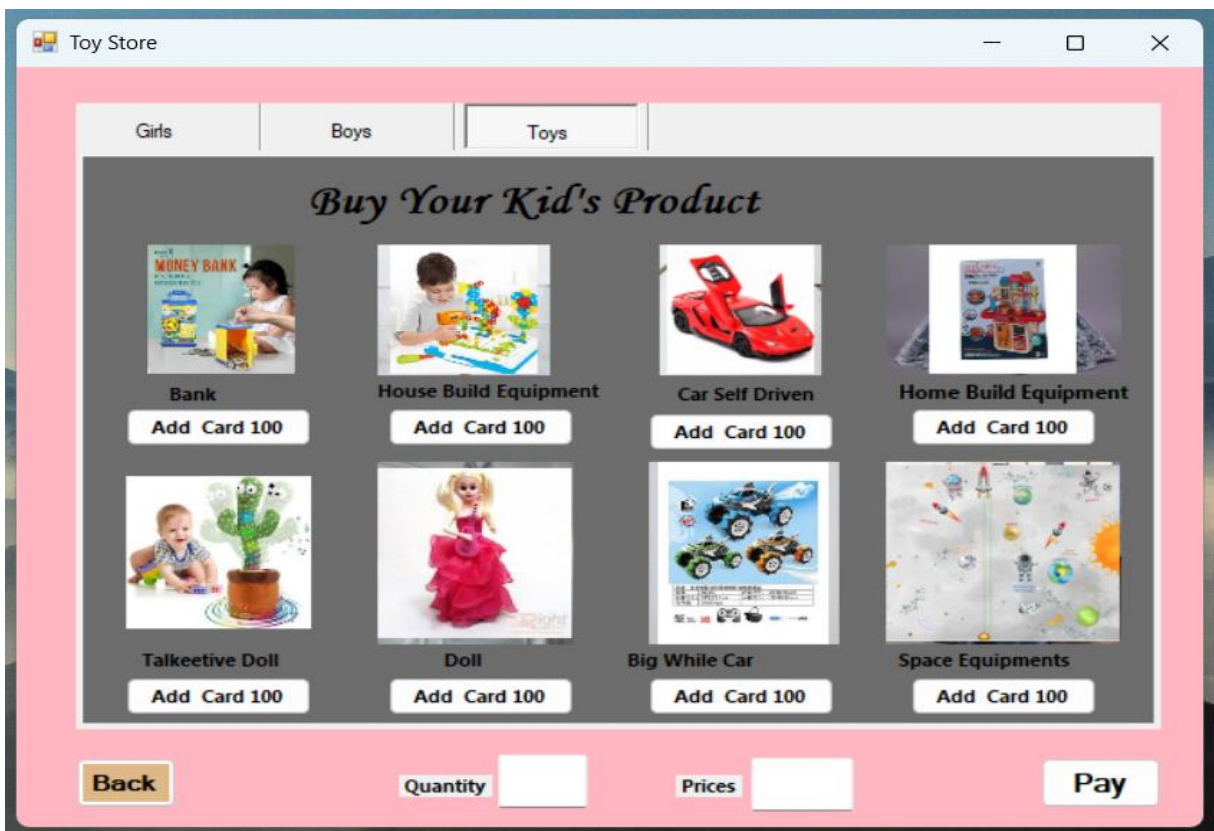
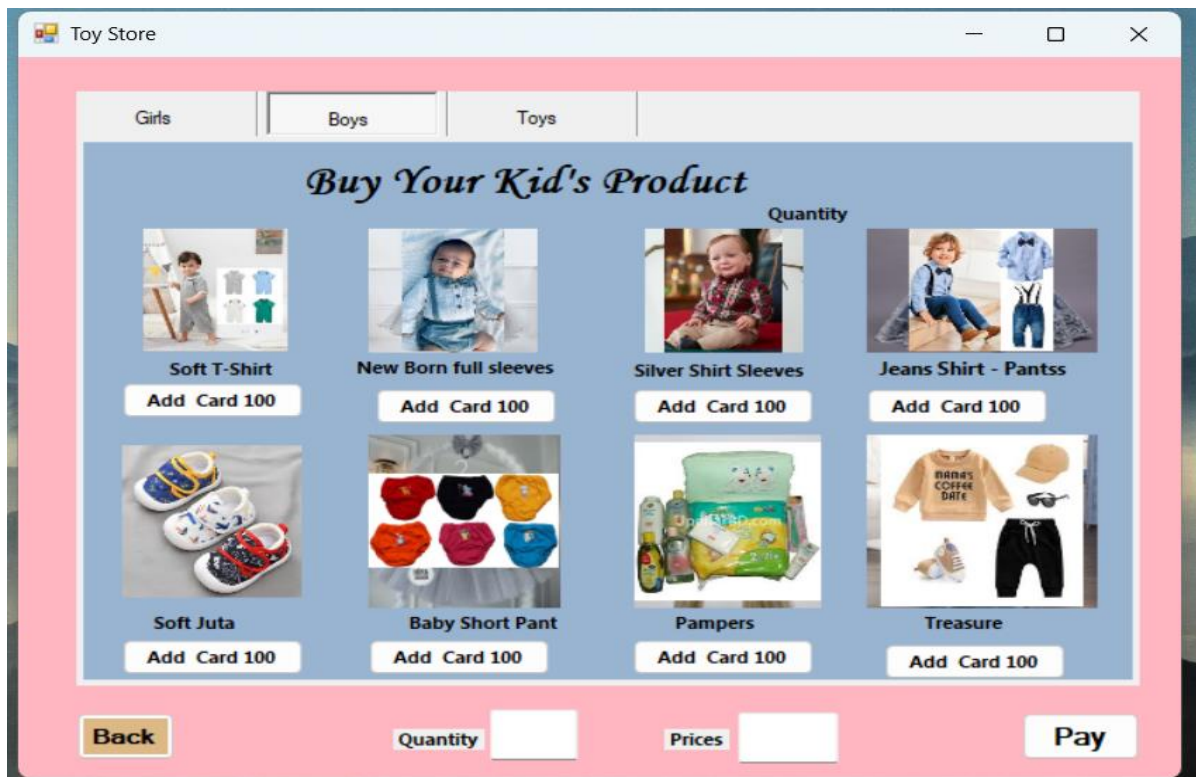
Add Card 100



ZAPAKA Blue Flower

Add Card 100

BackQuantityPricesPay



Payment Option

Back

Payment
19000 Taka

Choose one to proceed -

☒ Bkash

☐ Card

Confirm

Card Payment

Card

Number

Password

Back

Pay

Bkash Payment

bkash

Number

Password

Back **Pay**

Staff Page

Apply For Job

Login

Back

Applicant Form

STAFF HIRING

[Back](#)

First Name

Last Name

NID

DOB

Phone Num

Email

Gender ☐ **Male** ☐ **Female**

CV

[Submit](#)

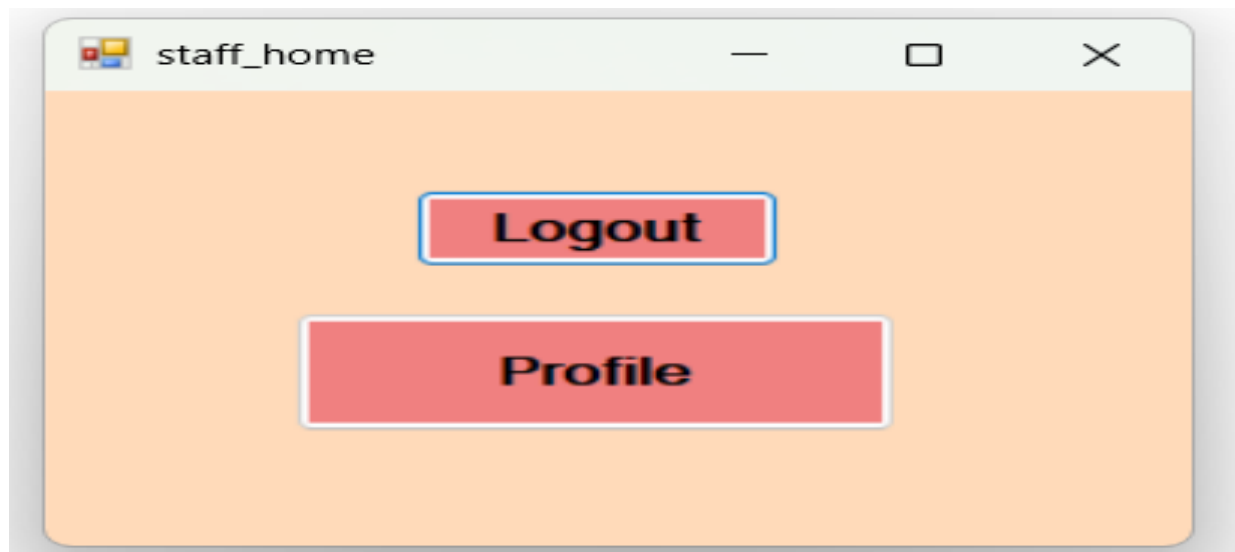
Staff Login Page

[Back](#)

ID :

Password :

[Login](#)

A screenshot of a web browser window titled "Staff Profile". The window has a dark blue background with a swirling pattern. It contains a form with various input fields and buttons. The form fields are labeled "ID", "New First Name", "New Last Name", "New Password", "New Nid", "New Email", "New Contact", "Job Title", "Salary", "Back", and "Update". The values entered in the fields are: ID: 3001, New First Name: Salma, New Last Name: Begum, New Password: 123, New Nid: 45564575634, New Email: jdksgfhj@gmail.com, New Contact: 01766554371, Job Title: Doctor, Salary: 20000.

Data Tables :

```
SELECT TOP 1000 [admin_username]  
            , [admin_pass]  
FROM [ChildCareDB].[dbo].[admin_table]
```

admin_username	admin_pass
aaa	aaa

```

SELECT TOP 1000 [guardian_id]
      ,[guardian_password]
      ,[guardian_f_name]
      ,[guardian_l_name]
      ,[guardian_nid]
      ,[guardian_phn_num]
      ,[guardian_email]
      ,[guardian_address]
      ,[guardian_emergency_num]
      ,[guardian_profession]
FROM [ChildCareDB].[dbo].[guardian_table]

```

	guardian_id	guardian_password	guardian_f_name	guardian_l_name	guardian_nid	guardian_phn_num	guardian_email	guardian_address	guardian_emergency_num	guardian_profession
1	1002	1234	Shamin	Yasser	345645674321	000000000	shaminyasser001@gmail.com	Bogra	01987654321	Student
2	1005	234	Sumi	Begum	112323	01719108734	sumi@gmail.com	Maitbag	01819108734	Doctor
3	1006	112	Nita	Jahan	227712	01577223147	nita@gmail.com	Mirpur	01568223148	Engineer
4	1007	dh843d49wb	Rahim	Rahman	368334563467	01976527891	rahim33945@gmail.com	Khulna	01976527891	Layer
5	1010	gie498hg90	Kamal	Hasan	358064322577	01987578945	kamal@gmail.com	Sylhet	01987578945	Farmer
6	1013	1111	Sakib	Yasser	948356	46457	sakib@gmail.com	afsdg	46457	sdgdfh

```

/***** Script for SelectTopNRows command fr
SELECT TOP 1000 [child_id]
      ,[child_f_name]
      ,[child_l_name]
      ,[child_dob]
      ,[child_gender]
      ,[child_blood_grp]
      ,[child_guardian_id]
      ,[child_assign_staff_id]
      ,[child_section_grp]
      ,[child_health_info]
FROM [ChildCareDB].[dbo].[child_table]

```

	child_id	child_f_name	child_l_name	child_dob	child_gender	child_blood_grp	child_guardian_id	child_assign_staff_id	child_section_grp	child_health_info
1	2001	Rahima	Alam	2015	Female	B+	1002	3001	PreSchool	
2	2002	Rahima	Begum	2015	Female	A-	1006	3004	NULL	NULL
3	2003	Kamrul	Hasan	2019	Male	O-	1005	3005	NULL	NULL
4	2004	Alia	Bhatun	2022	Female	O+	1014	NULL	NULL	NULL
5	2005	Sumana	Hoque	2019	Female	AB+	1015	NULL	NULL	NULL
6	2006	Jamal	Dou	2020	Male	B-	1016	NULL	NULL	NULL

```

SELECT TOP 1000 [applicant_id]
      ,[applicant_f_name]
      ,[applicant_l_name]
      ,[applicant_nid]
      ,[applicant_dob]
      ,[applicant_gender]
      ,[applicant_phn_num]
      ,[applicant_email]
      ,[applicant_cv_link]
FROM [ChildCareDB].[dbo].[applicant_table]

```

```

-- Script for SelectTopNRows command
SELECT TOP 1000 [staff_id]
      ,[staff_password]
      ,[staff_f_name]
      ,[staff_l_name]
      ,[staff_nid]
      ,[staff_dob]
      ,[staff_gender]
      ,[staff_phn_num]
      ,[staff_email]
      ,[staff_job_title]
      ,[staff_salary]
FROM [ChildCareDB].[dbo].[staff_table]

```

	staff_id	staff_password	staff_f_name	staff_l_name	staff_nid	staff_dob	staff_gender	staff_phn_num	staff_email	staff_job_title	staff_salary
1	3001	123	Salma	Begum	45564575634	1996	Female	01766554371	jdksgthj@gmail.com	Doctor	20000
2	3002	453	Nita	Ahmed	332145	1998	Female	01752534915	nita@gmail.com	Cleaner	10000
3	3003	229	Kamal	Mia	212315	1994	Male	01734523119	kamal@gmail.com	Security	15000
4	3004	107	Dipa	Saha	453216	1999	Female	0157683410	dipa@gmail.com	Caretaker	16000
5	3005	123456789	Akkae	Rahman	8430485406	13-11-1988	Male	01934567456	amarvai@gmail.com	Toddler	20000

```

/***** Script for SelectTopNRows command from
SELECT TOP 1000 [applicant_id]
      ,[applicant_f_name]
      ,[applicant_l_name]
      ,[applicant_nid]
      ,[applicant_dob]
      ,[applicant_gender]
      ,[applicant_phn_num]
      ,[applicant_email]
      ,[applicant_cv_link]
FROM [ChildCareDB].[dbo].[applicant_table]

```

	applicant...	applicant_f_na...	applicant_l_na...	applicant_nid	applicant_dob	applicant_gen...	applicant_phn_n...	applicant_email	applicant_cv_link
1	4013	Jamal	Rahman	3485473054	Thursday, January 23, 2025	Male	01968902345	jamal039@gmail.com	www.skfsdfksdf.com
2	4018	Kamal	Hossain	4453456135	2001	Male	01934534563	kamrul98@gmail.com	www.dsfgs.com
3	4021	Jamila	Khatun	4353467865	1999	Female	01734992124	jamila12@gmail.com	www.rfsghbdf.com
4	4022	Akkae	Rahman	8765432353	1989	Male	01345464346	akkas@gmail.com	www.eafsd.com
5	4026	Ali	Hossain	3538754564	1998	Male	01935345645	ali@gmail.com	www.eafsd.com

USE CASE DIAGRAM :

